

PRACTICAL - 1

(a)

```

harshika@siri: ~
ls
OS          excell.c  execv.c  execvp.c
OS-Skill    execlp    execve   fork_demo
OSLab       execlp.c  execve.c fork_demo.c
excell      execv     execvp
harshika@siri:~$

```

(b)

```

program2.c
~/OS/fork
1 #include <stdio.h>
2 #include <unistd.h>
3 int main()
4 {
5     fork();
6     printf("Hello\n");
7     return 0;
8 }
9
ion '1.0' not found
harshika@siri:~/OS/fork$ gedit program2.c
** (gedit:1370): WARNING **: 09:34:27.029: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:1370): WARNING **: 09:34:27.031: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OS/fork$ gcc program2.c -o program2
harshika@siri:~/OS/fork$ ./program2
Hello
Hello

```

(c)

```

excell1.c
~
1 #include <stdio.h>
2 #include <unistd.h>
3 int main()
4 {
5     printf("Before execl()\n");
6     execl("/bin/ls", "ls", "-l", NULL);
7     printf("This will not print if execl() succeeds.\n");
8     return 0;
9 }
10
harshika@siri:~$ gedit excell1.c
** (gedit:551): WARNING **: 09:03:56.145: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:551): WARNING **: 09:03:56.146: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc excell1.c -o excell
harshika@siri:~$ ./excell
Before execl()
total 48
drwxr-xr-x 3 harshika harshika 4096 Jul 30 09:31 OS
drwxr-xr-x 2 harshika harshika 4096 Jul 25 03:42 OSLab
-rwxr-xr-x 1 harshika harshika 15992 Jul 31 09:04 excell
-rw-r--r-- 1 harshika harshika 187 Jul 31 09:04 excell.c
-rwxr-xr-x 1 harshika harshika 16040 Jul 28 07:06 fork_demo
-rw-r--r-- 1 harshika harshika 176 Jul 28 07:06 fork_demo.c

```

(d)

```
Open  ▾  +  program5.c  ~./OS/fork  Save  ≡  x

1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/wait.h>
4 int main()
5 {
6     pid_t pid = fork();
7     if(pid == 0)
8     {
9         printf("Child Executing\n");
10    }
11    else
12    {
13        wait(NULL);
14        printf("Parent Executing\n");
15    }
16    return 0;
17 }
```

```
Parent PID = 3427
harshika@siri:~/OS/fork$ gedit program5.c

** (gedit:3440): WARNING **: 09:37:34.577: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:3440): WARNING **: 09:37:34.578: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OS/fork$ gcc program5.c -o program5
harshika@siri:~/OS/fork$ ./program5
Child Executing
Parent Executing
```

(e)

```
Open  ▾  +  program4.c  ~./OS/fork  Save  ≡  x

1 #include <stdio.h>
2 #include <unistd.h>
3 int main()
4 {
5     pid_t pid;
6     pid = fork();
7     if(pid == 0)
8     {
9         printf("Child PID = %d\n", getpid());
10        printf("Parent PID = %d\n", getppid());
11    }
12    else
13    {
14        printf("Parent PID = %d\n", getpid());
15        printf("Child PID = %d\n", pid);
16    }
17    return 0;
18 }
```

```
harshika@siri:~/OS/fork$ gedit program4.c

** (gedit:2849): WARNING **: 09:36:50.245: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:2849): WARNING **: 09:36:50.246: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~/OS/fork$ gcc program4.c -o program4
harshika@siri:~/OS/fork$ ./program4
Parent PID = 3427
Child PID = 3428
Child PID = 3428
Parent PID = 3427
```

Uname -

```
harshika@siri:~$ uname
Linux
harshika@siri:~$ |
```

[illegible]

```
Vmscape:                                     Not affected
```

```
harshika@siri:~$ lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	356.9M	1	disk	
sdb	8:16	0	159.4M	1	disk	
sdc	8:32	0	1G	0	disk	[SWAP]
sdd	8:48	0	1T	0	disk	/mnt/wslg/distro /

```
harshika@siri:~$ |
```

```
harshika@siri:~$ ps
  PID TTY          TIME CMD
  410 pts/0    00:00:00 bash
  760 pts/0    00:00:00 ps
harshika@siri:~$ |
```

Top -

```
760 pts/0    00:00:00 ps
harshika@siri:~$ top
top - 03:39:01 up 19 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running, 23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0
MiB Mem :  3764.1 total,  3248.4 free,   406.8 used,   183.1 buff/cache
MiB Swap:  1024.0 total,  1024.0 free,    0.0 used.  3357.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
    1 root        20   0   24512   15472   11556 S   0.0   0.4   0:02.36
    2 root        20   0    3180    2204    2072 S   0.0   0.1   0:00.10
    6 root        20   0    3180    2048    1940 S   0.0   0.1   0:00.00
   52 root        19  -1   50396   16700   15436 S   0.0   0.4   0:00.62
   91 systemd+    20   0   22416   14456   11964 S   0.0   0.4   0:00.25
   93 root        20   0   35240   12212    9256 S   0.0   0.3   0:01.47
  144 root        20   0    2888    1948    1816 S   0.0   0.1   0:00.09
  145 root        20   0    4472    3124    2876 S   0.0   0.1   0:00.01
  146 message+   20   0    8816    5368    4704 S   0.0   0.1   0:00.14
  155 root        20   0   44096   29888   14748 S   0.0   0.8   0:00.59
  161 root        20   0   18440    9420    8176 S   0.0   0.2   0:00.20
  169 syslog     20   0  220548    5588    4364 S   0.0   0.1   0:00.15
  233 root        20   0    5492    2696    2508 S   0.0   0.1   0:00.00
  270 root        20   0  123460   32520   17912 S   0.0   0.8   0:00.57
  274 _chrony     20   0   23688   11216    7228 S   0.0   0.3   0:00.18
  298 _chrony     20   0   12096    2436    1796 S   0.0   0.1   0:00.01
  407 root        20   0    3196    1112     980 S   0.0   0.0   0:00.00
  409 root        20   0    3196    1248    1108 S   0.0   0.0   0:00.05
  410 harshika   20   0    6268    5528    3856 S   0.0   0.1   0:00.16
  411 root        20   0    8644    5552    4712 S   0.0   0.1   0:00.02
  456 harshika   20   0   22756   13432   10972 S   0.0   0.3   0:00.23
  458 harshika   20   0   22916    4100    2096 S   0.0   0.1   0:00.00
  486 harshika   20   0    6136    5408    3852 S   0.0   0.1   0:00.06
  772 harshika   20   0    8296    6080    3900 R   0.0   0.2   0:00.02
```